IV Bolus Practice 4

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

1a 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. a. What is the expected initial concentration of the drug in the plasma?

1b 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. b. What percent of the dose is eliminated in 24 hours?

1c 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. c. What is the expected plasma drug concentration (C_p) at 12 hours after administration of the dose?

1d 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. d. If the dose were doubled, what is the expected change in the half-life of elimination?

A. The half-life increases, but by less than double

B. No change; the half-life is constant for a first-order process

C. The half-life doubles

D. The half-life halves

1e 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. e. If the dose were doubled, what is the expected change in the clearance?

A. Clearance halves

B. No change; clearance is constant for a first-order process

C. Clearance doubles

D. Clearance increases by a factor of 4

1f 1. A drug that follows one-compartment, first-order elimination has an elimination half-life of 6 hours and apparent volume of distribution V_D of 400 mL/kg has been administered as a single 600-mg dose is given to an adult female patient (62 kg) by rapid IV injection. f. If the dose were doubled, what is the expected change in the initial plasma concentration?

A. No change

B. It increases by a factor of 4

C. It halves

D. It increases by a factor of 2

2a 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

a. What is the elimination half-life of this drug?

2b 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

b. What is the initial plasma concentration of this drug?

2c 2. The pharmacokinetics of an antibiotic administered by IV bolus injection is best described by the equation below. (Concentration is given in mcg/mL and time in hours)

$C_p = 18.35e^{-4.8t} + 12.63e^{-0.09t}$

c. What is the concentration of drug in the plasma 4 hours after administration of the dose?